

Fingers Movement an XAI-centered Data Analysis

FINGER MOVEMENT DATASET

Each input is a multivariate time series (EEG signals over time from multiple sensors), and the task is to classify the intended movement.

Characteristic	Description
Samples	~316 training / ~100 test
Input Shape	(50, 28) — 50 time steps, 28 features (channels)
Each Sample	A short window of EEG signal from 28 electrodes
Tasks	Binary classification: Left vs Right hand movement intention
Labels	Class 0 = Left hand movement Class 1 = Right hand movement

FINGER MOVEMENT DATASET

MODEL METRICS

Model Used : LSTM
Model Accuracy - 0.61

Threshold - 0.5

Class	Precision	Recall	F1-Score
0 (Left)	0.61	0.57	0.59
1 (Right)	0.61	0.65	0.63

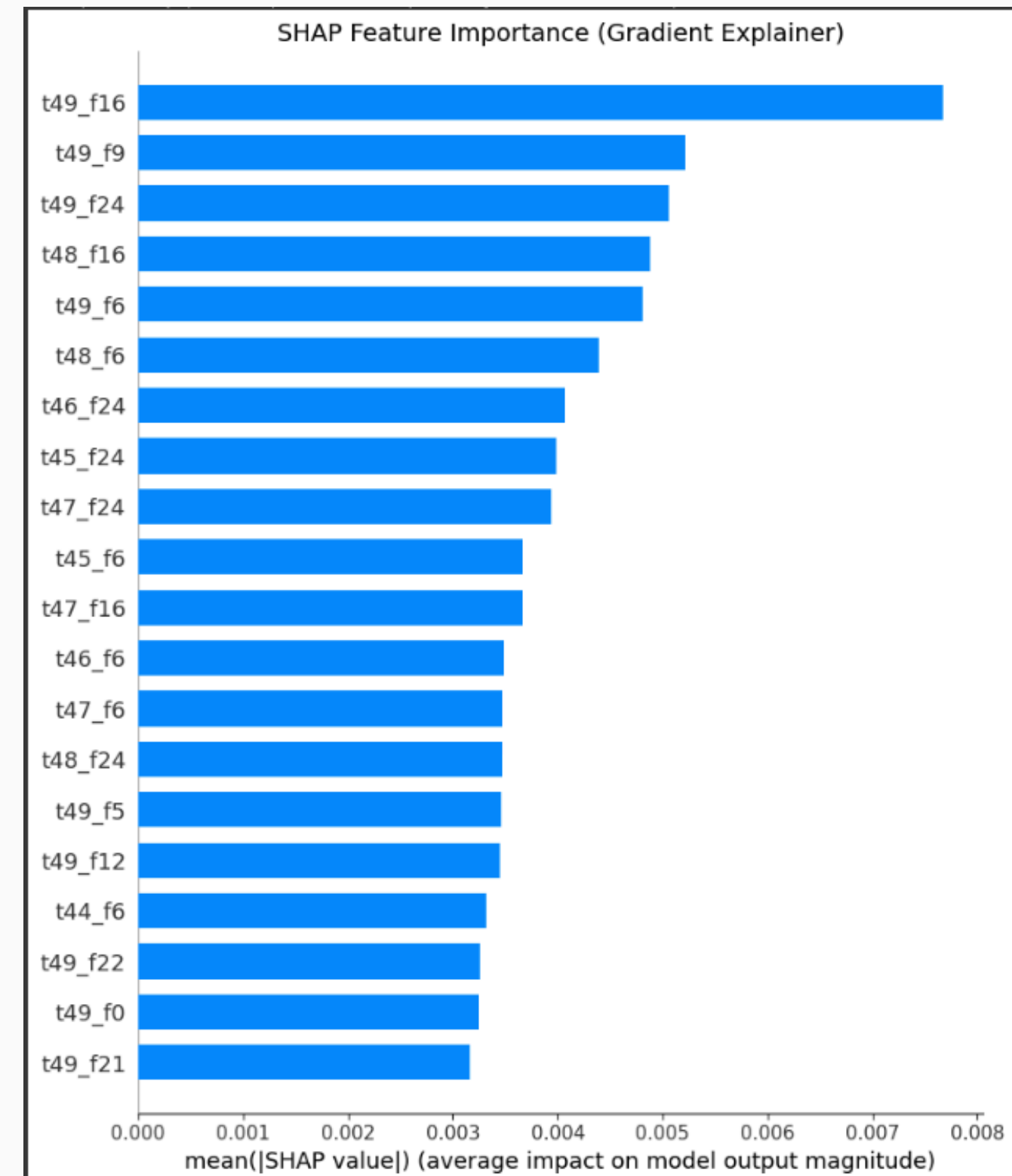
SHAP Global Importance

TOP N FEATURES

t49_f16 is the most influential feature for the model's predictions, suggesting that the signal recorded at time step 49 on channel/feature 16 is crucial for distinguishing finger movement classes.

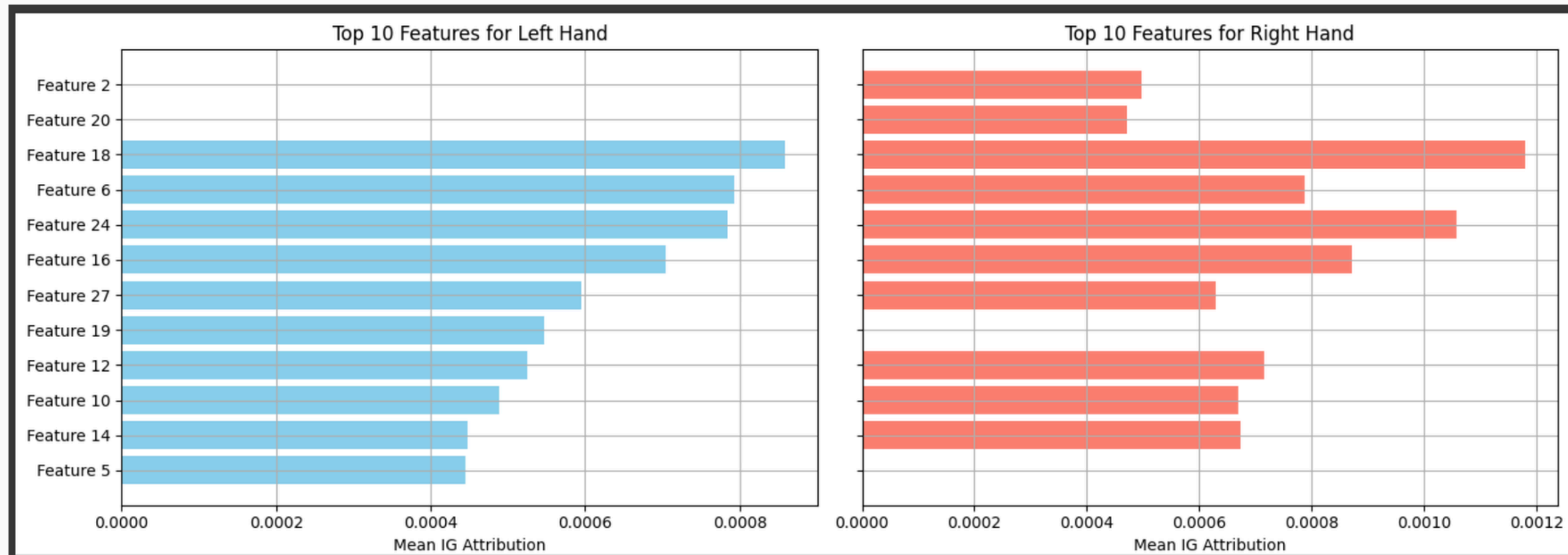
Features from later time steps (e.g., t49) and certain channels/features (f16, f9, f24, etc.) dominate the top ranks. This hints that the EEG activity towards the end of the measurement window, and in specific channels, carries key information for the classification task.

Multiple features from similar time points (t49, t45, t47, etc.) and channels are prominent, suggesting these intervals and sensors consistently hold discriminative patterns.



INTEGRATED GRADIENTS (Global Explanation)

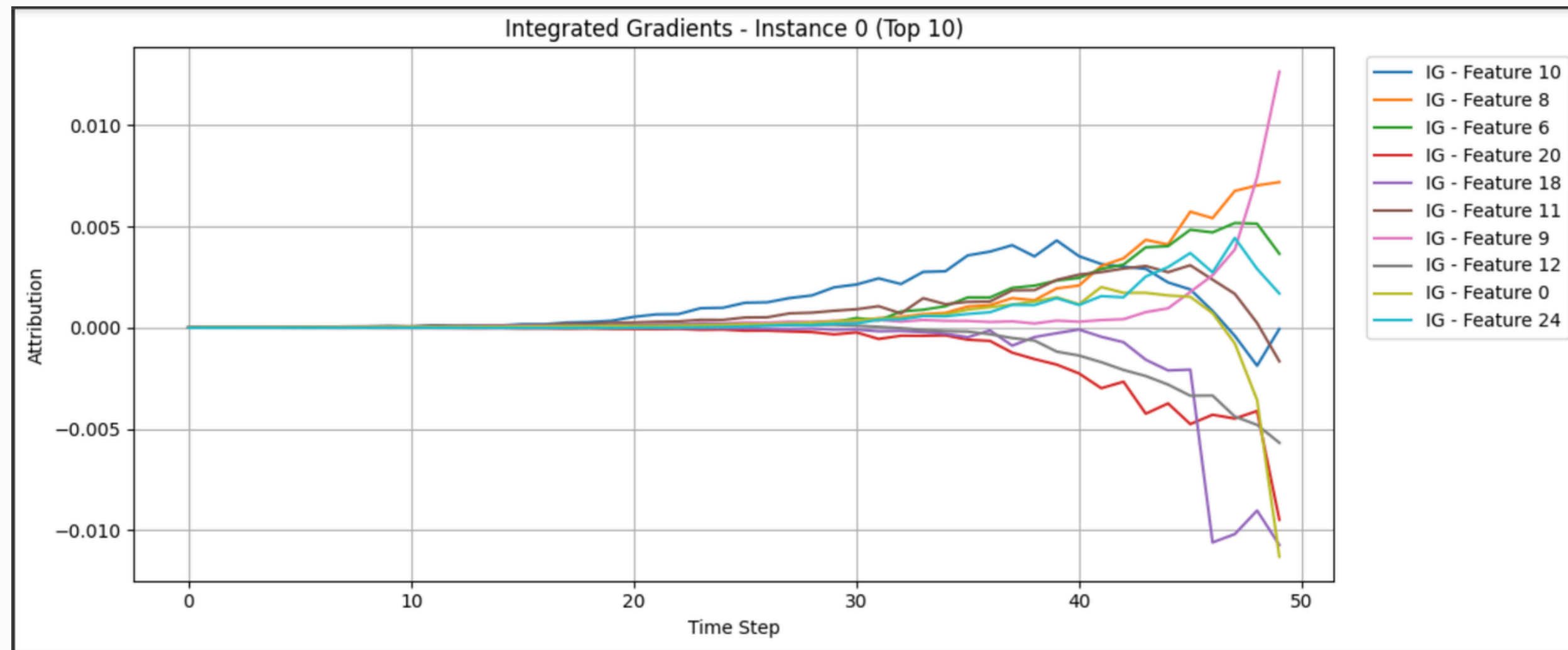
TOP N FEATURES BY CLASS



The feature attribution analysis shows that the most important EEG features for distinguishing left and right hand finger movements are partially distinct, with some overlap.

INTEGRATED GRADIENTS (Local Explanation)

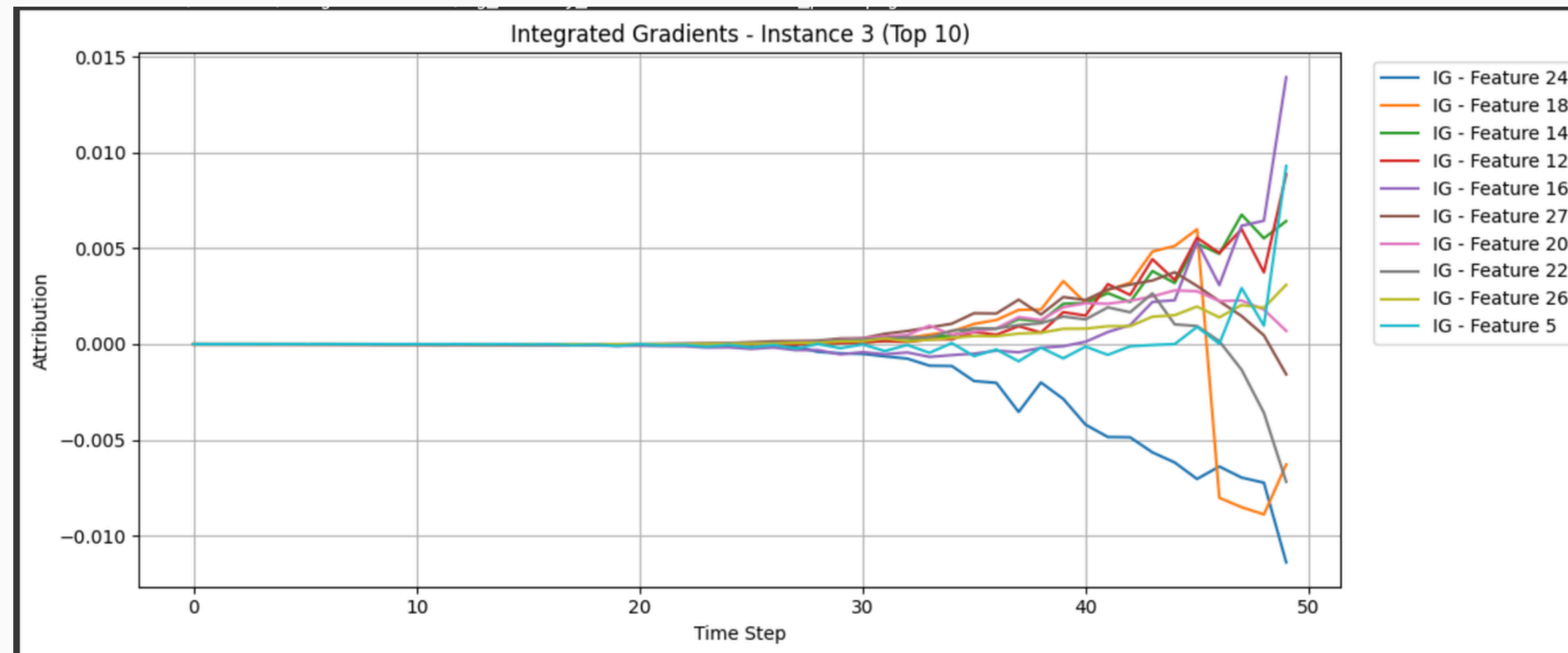
Example 1: (Actual = 1, Predicted = 1)



For most features, attribution gradually diverges after time step 20–30. This means the latter part of the EEG signal is particularly influential for this prediction.

INTEGRATED GRADIENTS (Local Explanation)

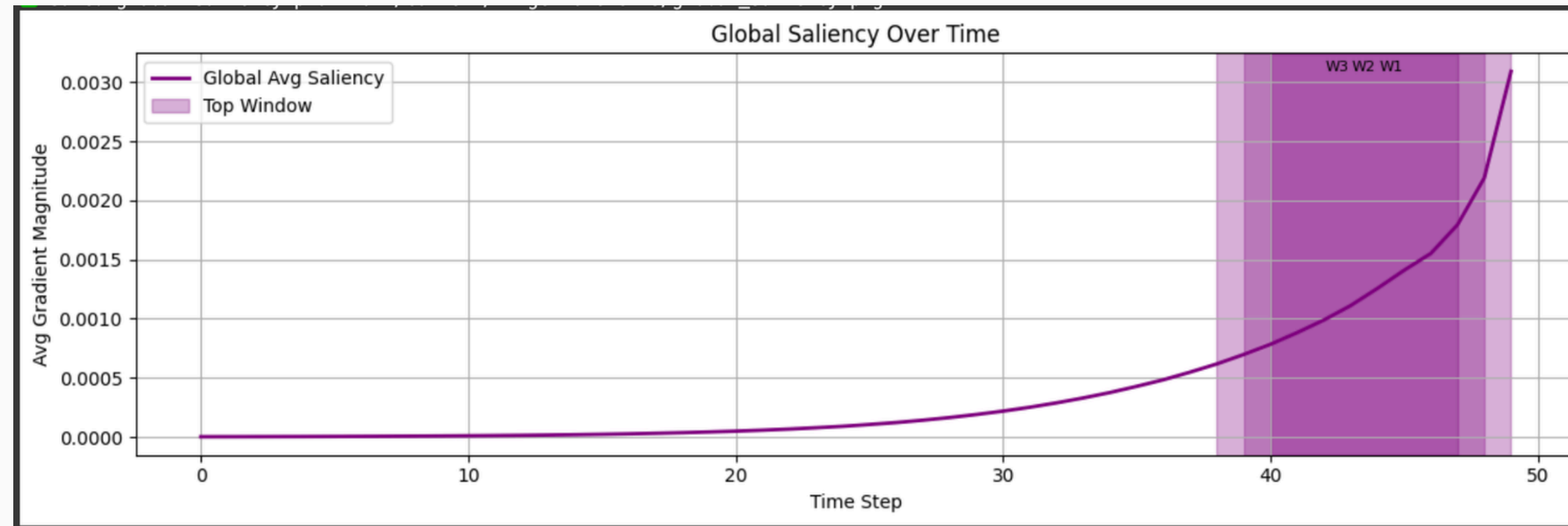
Example 2: (Actual = 0, Predicted = 1)



Most features show only minor attribution until after time step 30–35, where the curves start diverging.

The net attribution at the end of the sequence is positive for most features, reflecting that the model was ultimately confident in its prediction.

SALIENCY MAPS (Global Explanations)

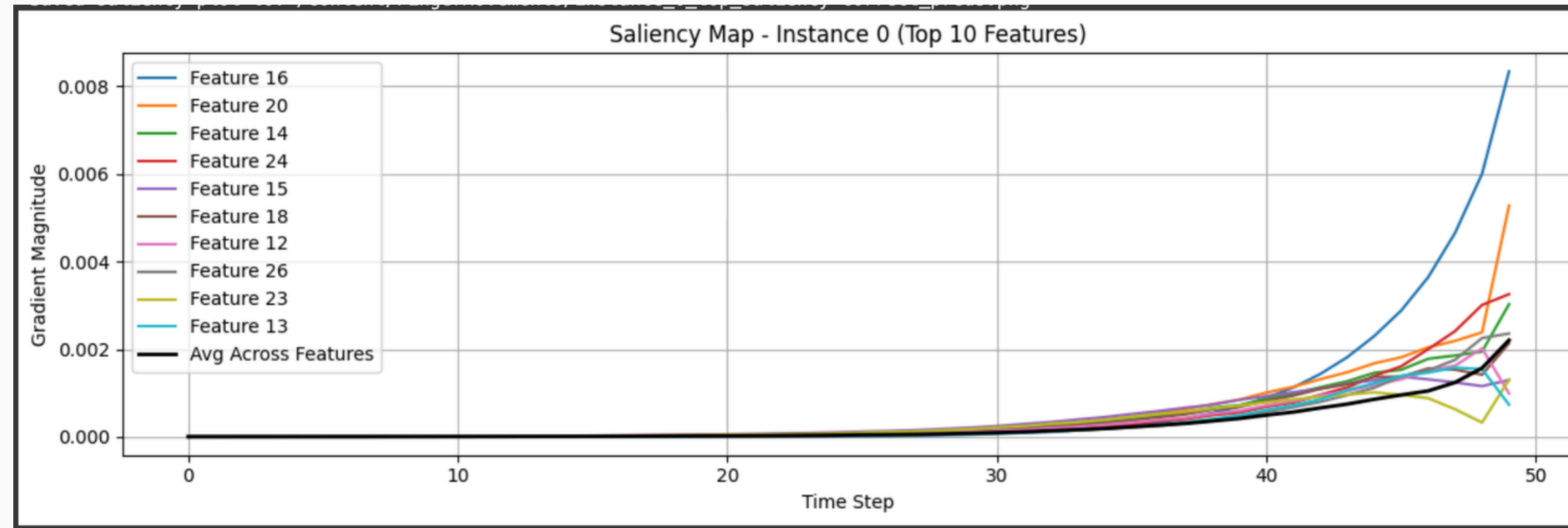


The plot shows the average saliency (gradient magnitude) across all features and instances, plotted over each time step (0–50).

The **highlighted window (t39–t50)** shows where the model’s sensitivity is maximal. This means **the model relies most heavily on the signal in these final time steps** when making predictions.

SALIENCY MAPS (Local Explanations)

Example 1: (Actual = 1, Predicted = 1)



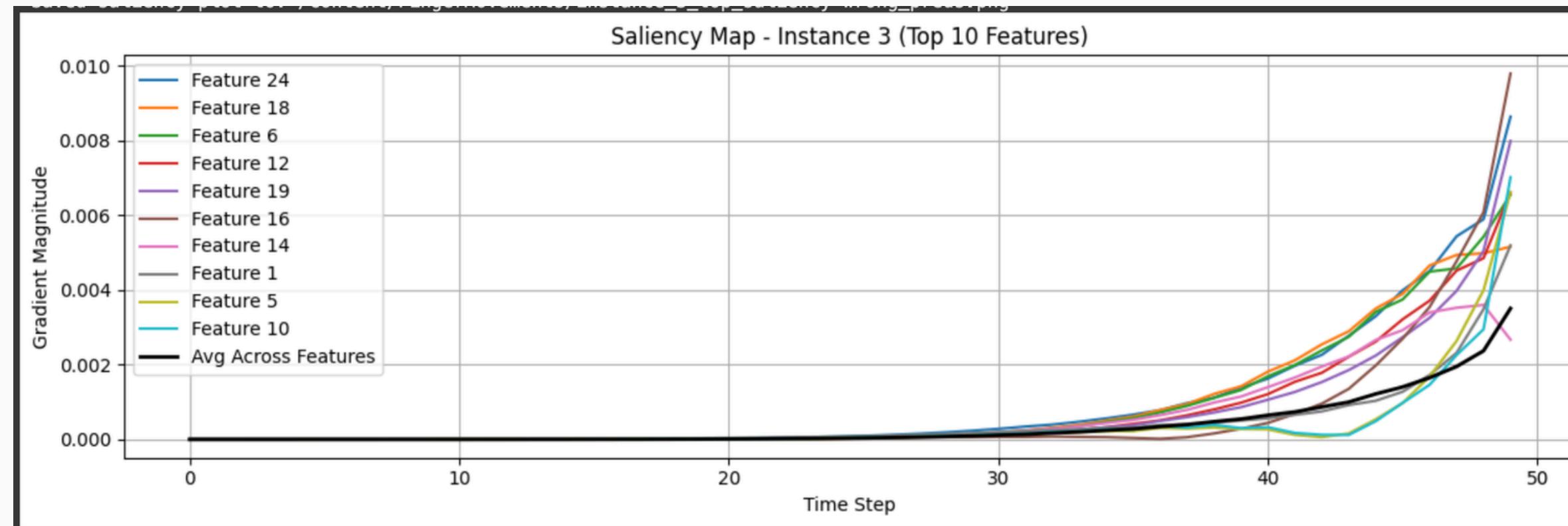
The largest peaks are all **at the very end of the sequence (t=50)**, confirming that the model's decision for this instance relies heavily on the final portion of the EEG data.

Feature 16 shows the highest peak at the final time step, followed by Features 14 and 24, etc.

This late-window focus may reflect a physiological signal associated with the execution of finger movements or the window right before/after movement onset.

SALIENCY MAPS (Local Explanations)

Example 2: (Actual = 0, Predicted = 1)



Strongest model sensitivity occurs in the final 10 time steps, especially at $t=50$, where most features peak.

This instance's pattern mirrors what pattern observed in Instance 0 and in the global saliency analysis: **the model relies most heavily on the end of the EEG window for prediction**, with certain features dominating.

The **average saliency (black line)** increases at the end, confirming that model sensitivity is not just driven by a single feature but is a general property of the late sequence

Conclusion

Finger Movements

- Model explanations show that key EEG features for classifying finger movements are concentrated at the end of each sequence, aligning with expected neural patterns during movement.
- Only a subset of channels contribute most to predictions, matching neuroscience insights on motor cortex activity.
- These findings support the validity of our model and suggest that late-window EEG signals carry the most discriminative information for finger movement classification.